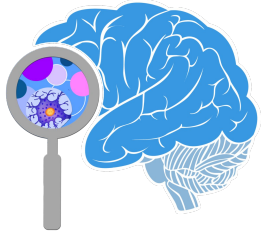


Aerobic respiration

Lesson objectives:

- To state the word equation for aerobic respiration.
- To explain why respiration is important in living organisms.



Year 7 recall!

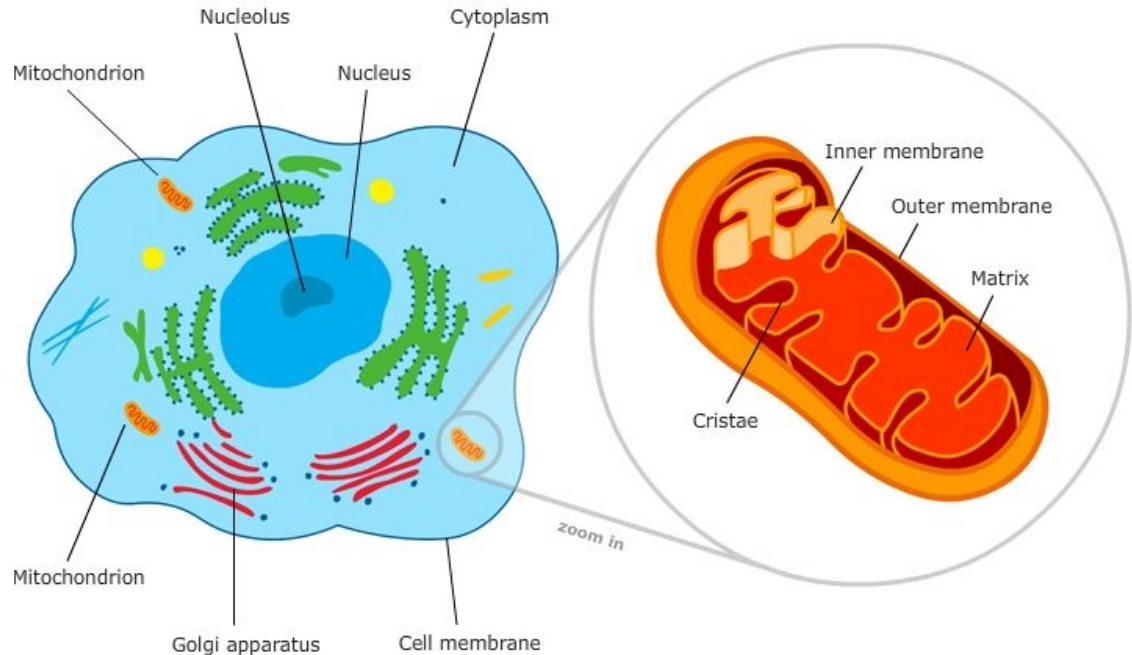
Draw and label an **animal cell**.

Respiration

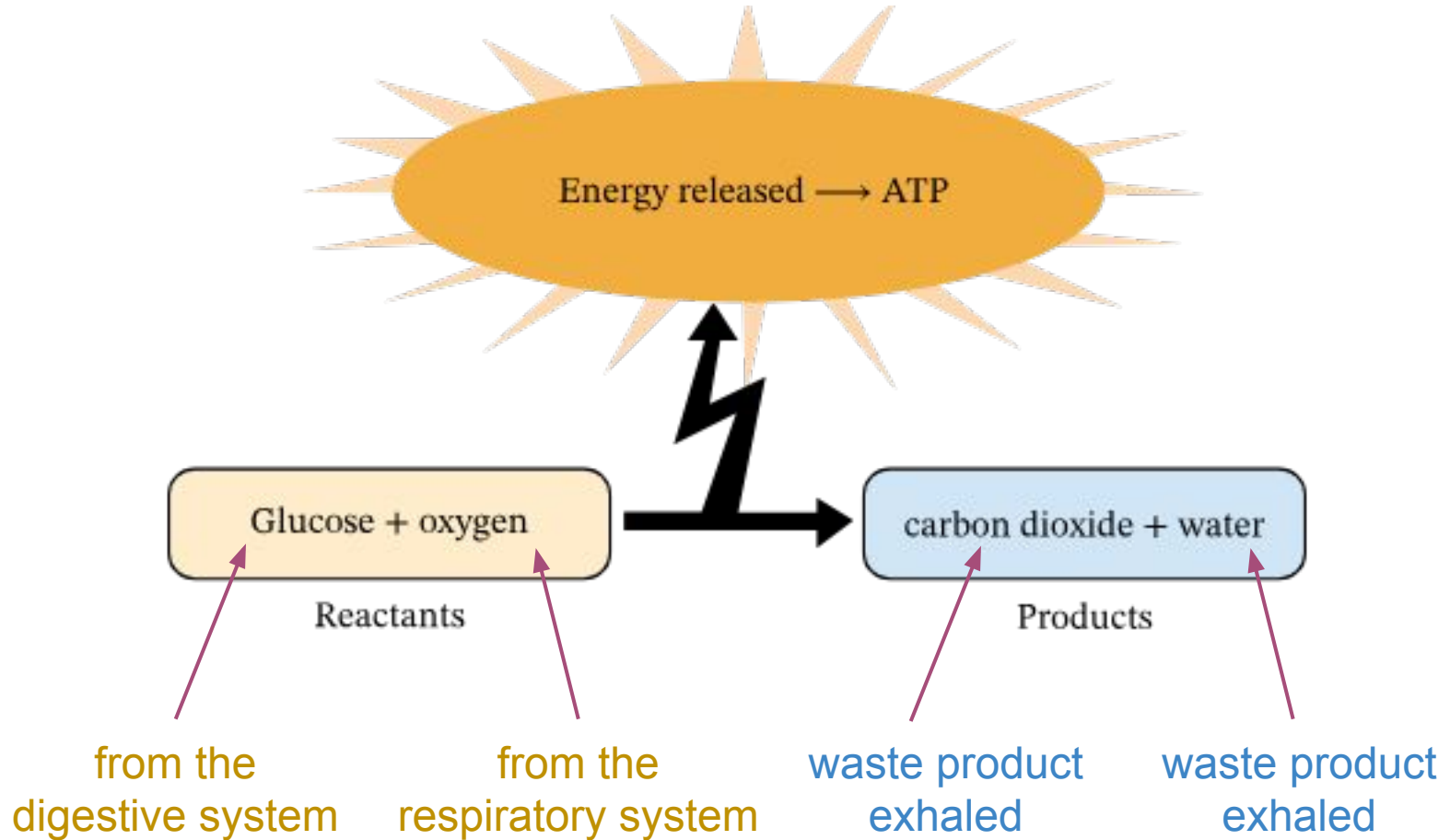
Respiration is a **chemical reaction** in which **energy is released** from food substances. Glucose is the substance that animals use for energy.

Remember: energy cannot be made or destroyed, *only transferred*.

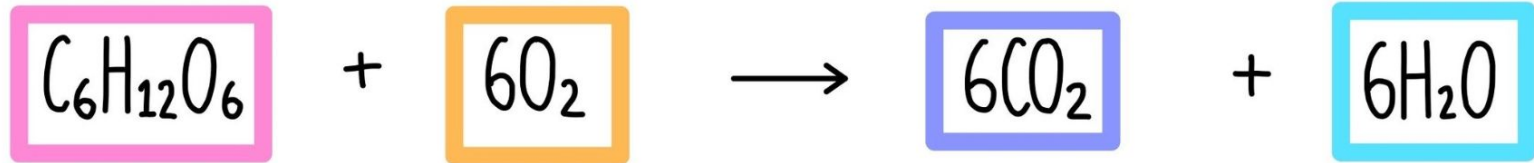
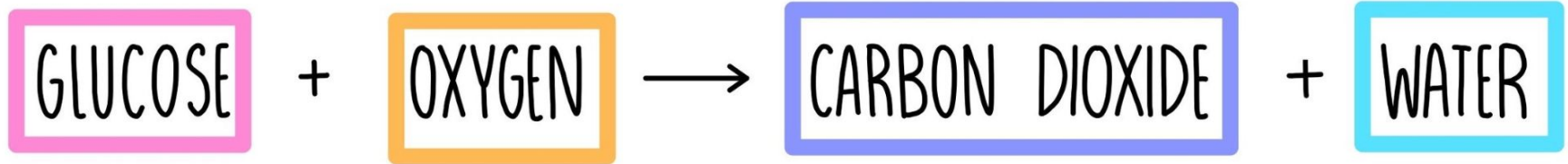
It takes place in the **mitochondria**.



Aerobic respiration



Aerobic respiration



Use the equation for aerobic respiration to **explain** why the air you breathe out contains more carbon dioxide than the air that you breathe in.

What is breathing?

The breathing system is used by the body to **take in** the **oxygen** needed for **respiration**.

It is also use to **remove** the **waste products** of respiration (**carbon dioxide and water**).

Copy and complete the sentences below:



Breathing involves the lungs and allows our bodies to draw in air from the outside environment. We breathe in and out through our mouth and our _____.

Respiration is a chemical _____ that takes place in all of our _____. It is a reaction between glucose and _____ to make water, carbon dioxide and release _____.

What is breathing?

The breathing system is used by the body to **take in** the **oxygen** needed for **respiration**.

It is also use to **remove** the **waste product** of respiration (**carbon dioxide and water**).

Copy and complete the sentences below:



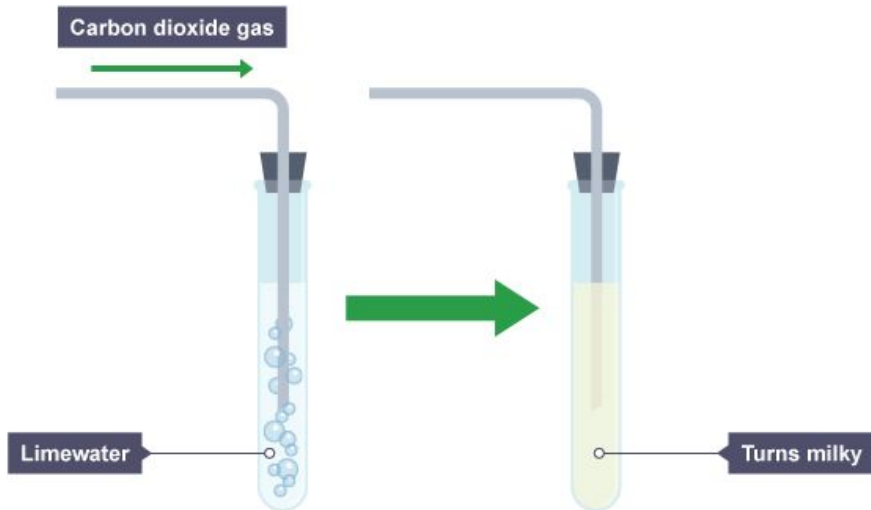
Breathing involves the lungs and allows our bodies to draw in air from the outside environment. We breathe in and out through our mouth and our **nose**.

Respiration is a chemical **reaction** that takes place in all of our **cells**. It is a reaction between glucose and **oxygen** to make water, carbon dioxide and release **energy**.

Investigating respiration

Respiration by living organisms can be investigated by carrying out experiments to show the production of carbon dioxide and heat.

Limewater can be used to detect carbon dioxide.



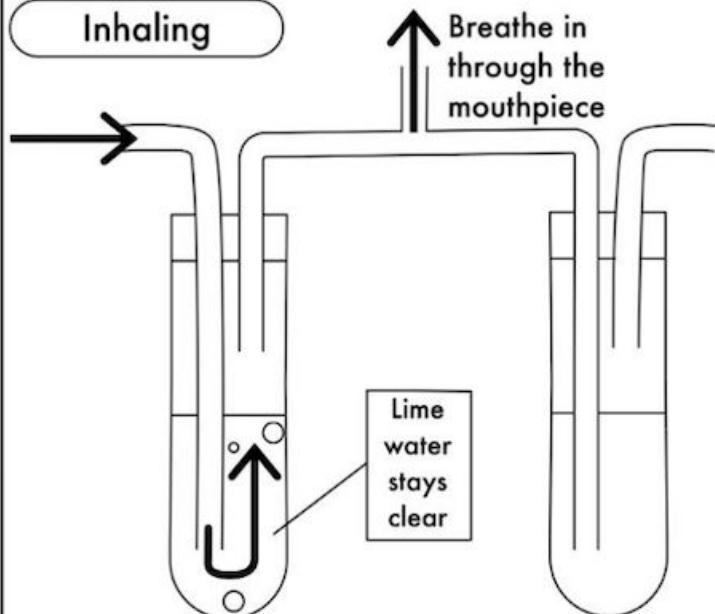
If carbon dioxide is bubbled through limewater then it **turns from clear to cloudy/milky** in colour.

HUFF AND PUFF

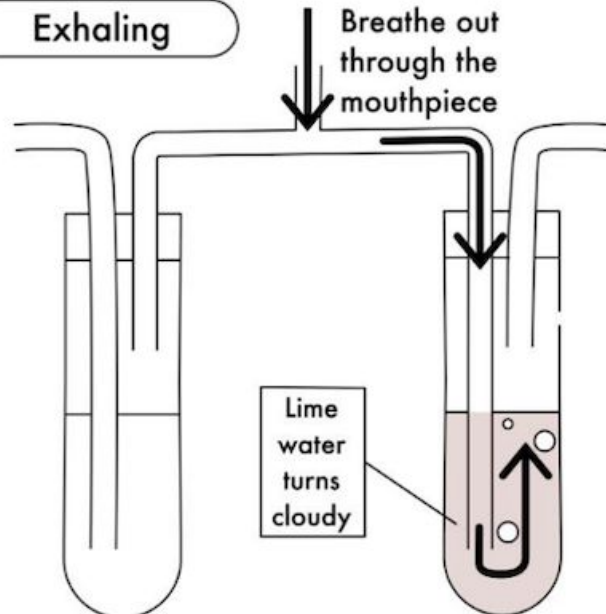
4

The huff and puff apparatus can be used to show the difference between the composition of inhaled and exhaled air. Both tubes contain lime water which turns cloudy or milky with increased levels of carbon dioxide.

Inhaling



Exhaling



Explain why the limewater on the left side stays clear

.....

.....

.....

.....

.....

Explain why the limewater on the right side turns cloudy

.....

.....

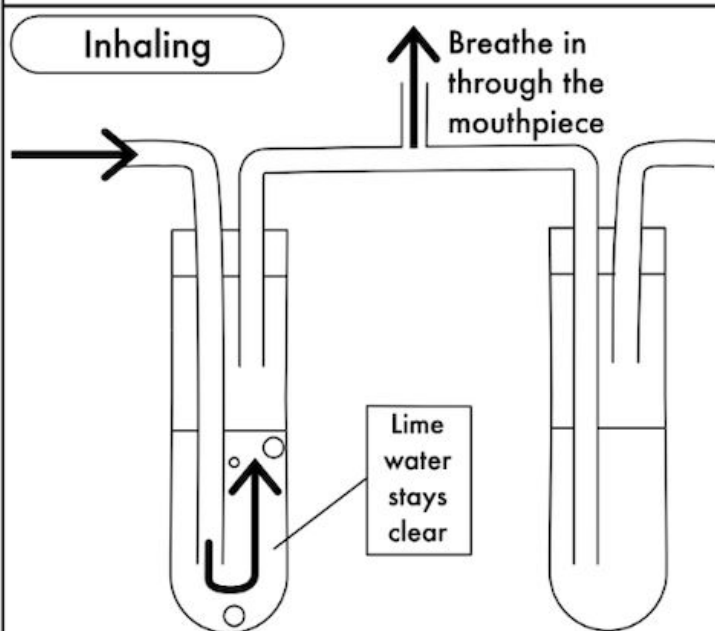
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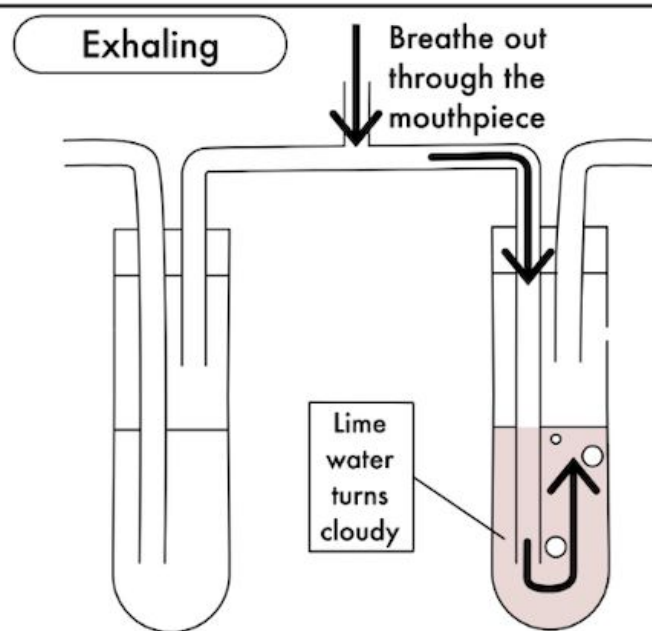
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HUFF AND PUFF

The huff and puff apparatus can be used to show the difference between the composition of inhaled and exhaled air. Both tubes contain lime water which turns cloudy or milky with increased levels of carbon dioxide.



Explain why the limewater on the left side stays clear
The inhaled air passes through the limewater and as it has a relatively low carbon dioxide concentration (0.04%) it stays clear.

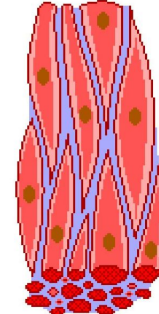
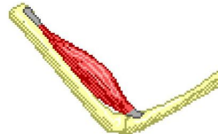
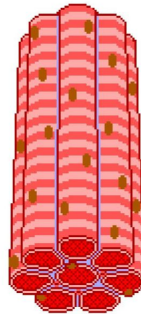
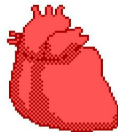
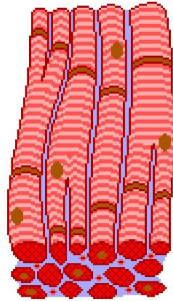
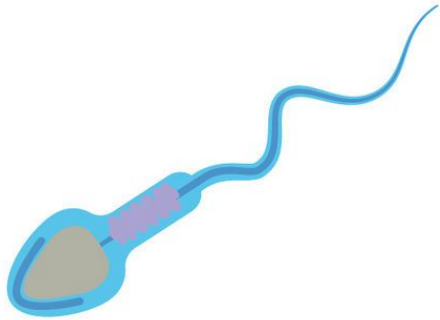


Explain why the limewater on the right side turns cloudy
The exhaled air passes through the limewater and as it has a relatively high carbon dioxide concentration (4%) it turns cloudy. This is because carbon dioxide has been added to the air we exhale from respiration in our cells.

Where is energy needed?

Respiration takes place in all cells of the body however some tissues/cells/organs require more energy than others.

Which cells have the most mitochondria in them?



Biology

AEROBIC RESPIRATION



1. What is the process that allows oxygen to move into cells and carbon dioxide to leave?
2. What does aerobic mean?
3. What is the chemical formula for glucose?
4. What are the products of aerobic respiration?
5. True or false - Respiration occurs in all body cells

1. What is the process that allows oxygen to move into cells and carbon dioxide to leave?

Diffusion

2. What does aerobic mean?

'With oxygen'

3. What is the chemical formula for glucose?



4. What are the products of aerobic respiration?

Carbon dioxide + Water + Energy

5. True or false - Respiration occurs in all body cells

True